RO2 - Lecture 5
3D Gyroscope

we focus on students

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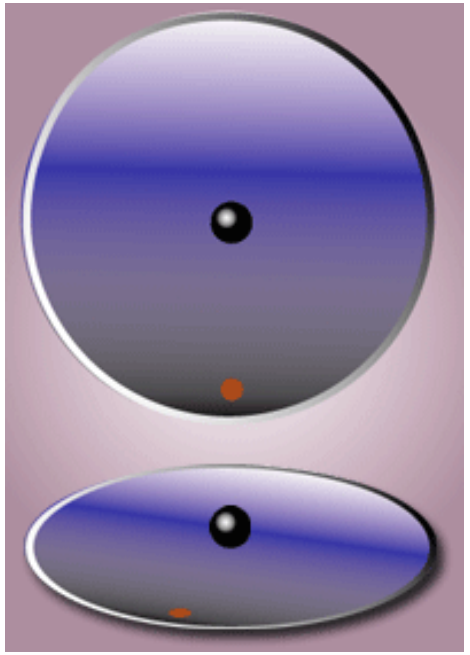
3D gyroscope data

- 3D gyroscopes measure the angular rate ω in deg/s (or rad/s) about the *x*, *y*, and *z*-axis
- MEMS (Micro-Electro-Mechanical Systems) gyroscopes mostly rely on the the *Coriolis Effect*
- Coriolis Effect refers to the apparent force that is exerted on a mass moving in a rotating system



source

- ball moves in a straight line
- the observer (red dot) in the rotating frame of reference sees the object as following a curved path due to the Coriolis effect
- Coriolis Force described as $F_C = 2m(\boldsymbol{v}_{rel} \times \boldsymbol{\omega})$

  where:

  - $\boldsymbol{v}_{rel}$ is the ball's velocity relative to the rotating reference frame (i.e. the frame of the observer rotating with the disk)
  - $\boldsymbol{\omega}$ is the angular velocity vector of the rotating frame (the disk)
  - m is the mass of the object being acted on – here: the mass of the ball

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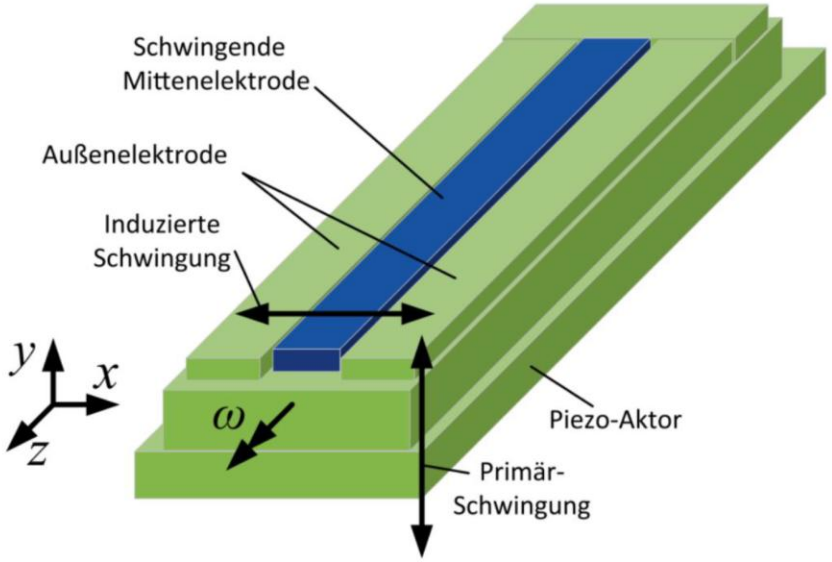
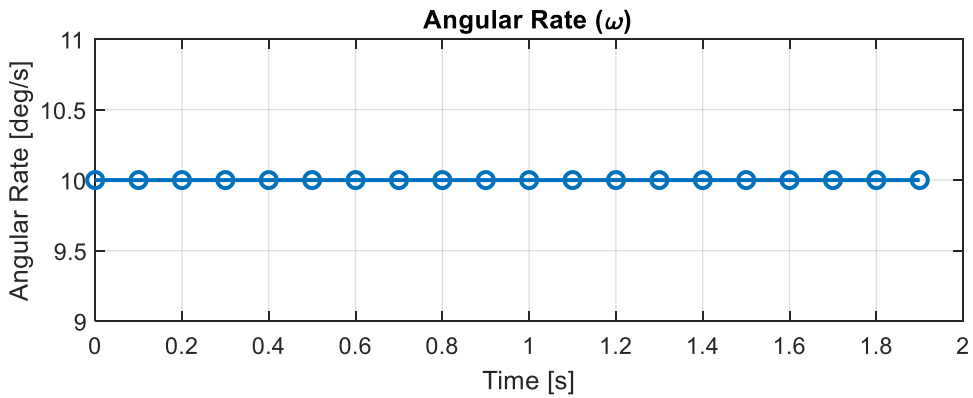
- MEMS (Micro-Electro-Mechanical Systems) gyroscopes mostly rely on the the *Coriolis Effect*
- Coriolis Effect refers to the apparent force that is exerted on a mass moving in a rotating system
- This effect is used in MEMS sensors as follows:



- Piezo actuator initiates primary oscillation in vertical direction (here along *y*-axis)
- When the device rotates about the *z*-axis, the Coriolis force induces a secondary displacement of the middle electrode in the *x*-direction
- The resulting differential capacitance change at the outer electrodes is measured and used to compute the angular rate ω

3D gyroscope data

- By integrating the data, the orientation angle of each axis can be determined
- Discrete (sampled) data with measurements $\omega(k)$ for $k = 0,1, \ldots$ as sample times with period T_s



Constant angular rate of 10 deg/s,
Sample Period of $T_s = 0.1\mathrm{s}$

What is the angle after 10 samples?

$$10 \cdot T_s = 10 \cdot 0.1\mathrm{s} = 1\mathrm{s}$$

$$1\mathrm{s} \cdot 10\,\frac{\mathrm{deg}}{\mathrm{s}} = 10\mathrm{deg}$$

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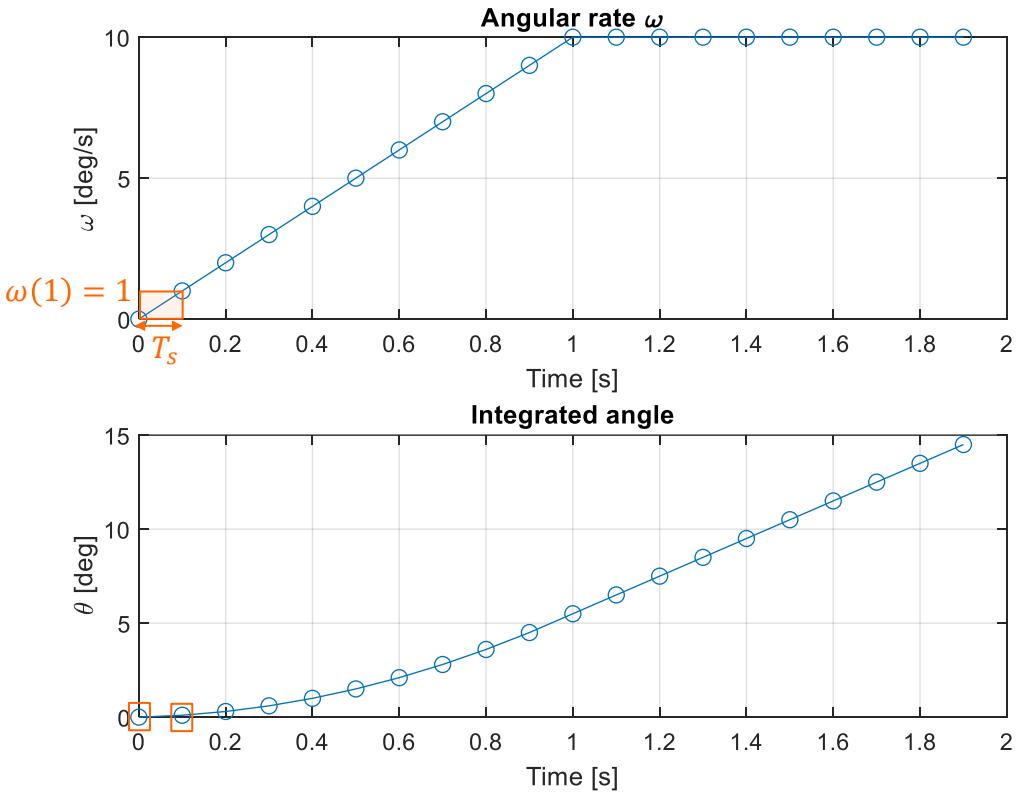

How to determine the angle values for each sample, starting at k = 0?

- $k = 0$ → Initial value $\theta(0) = 0°$
- $k = 1$

→ $\theta(1) = 0° + T_s\omega(1) = T_s\, 1\frac{°}{s} = 0.1s \cdot 1\frac{°}{s} = 0.1°$

3D gyroscope data

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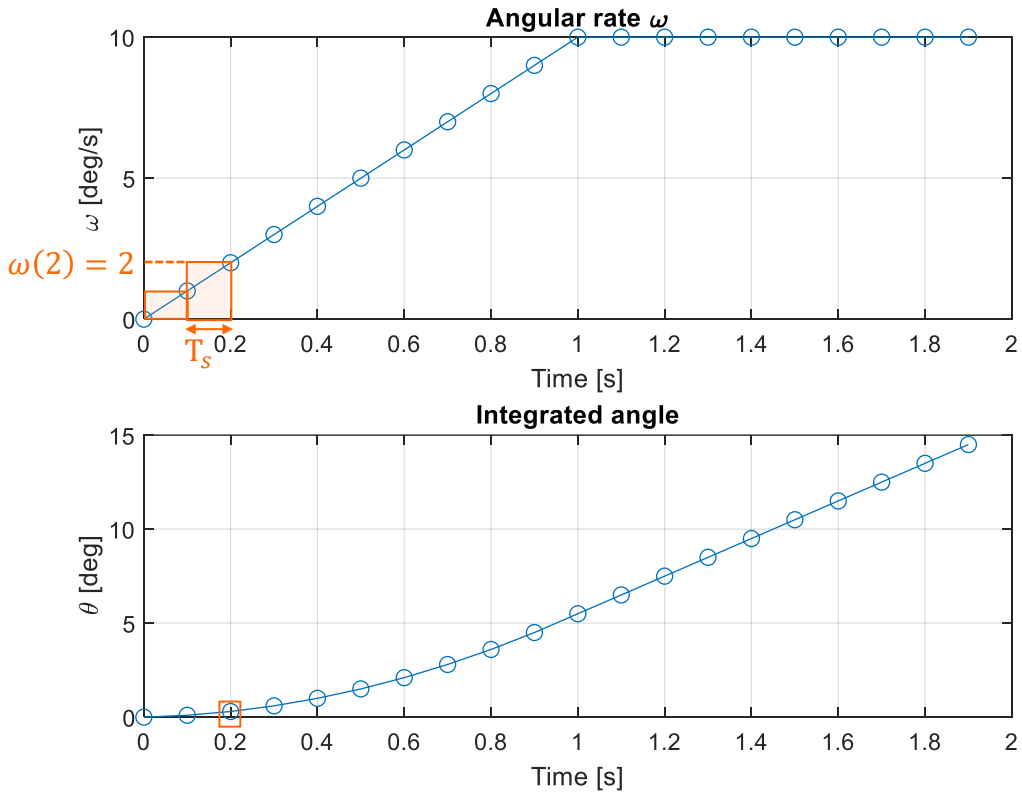

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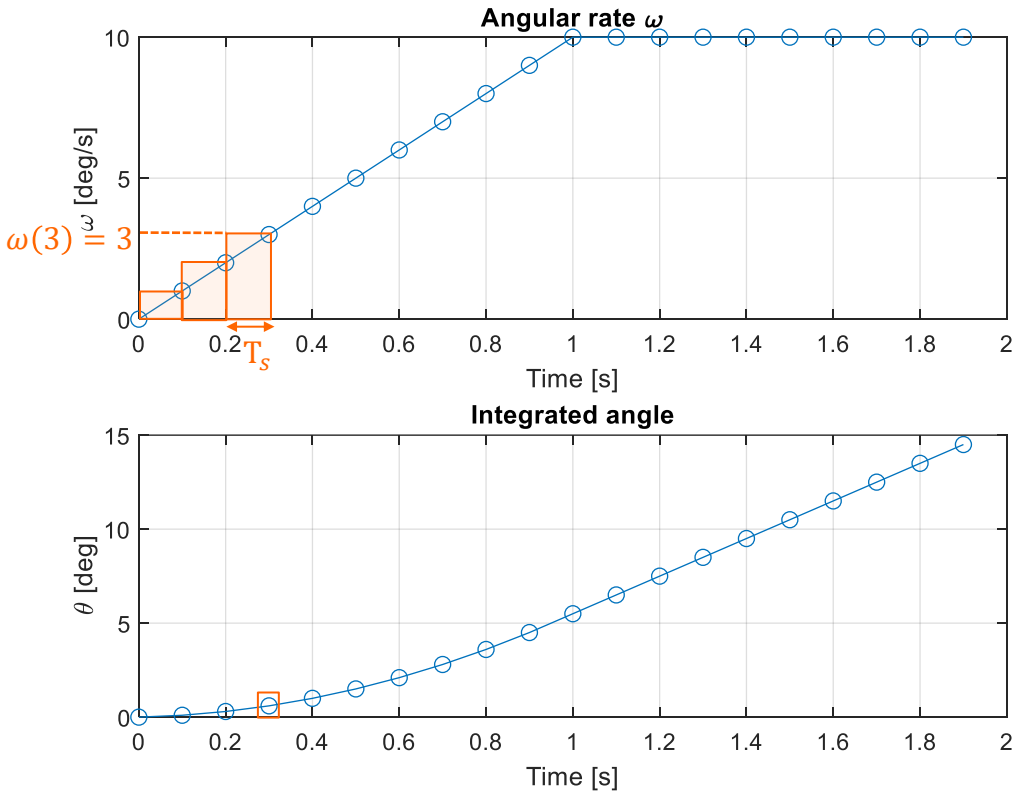
- $k = 2$

→ $\theta(2) = \theta(1) + T_s\omega(2) = 0.1° + 0.1s \cdot 2\frac{°}{s} = 0.3°$

3D gyroscope data

- By integrating the data, the orientation angle of each axis can be determined
- Discrete (sampled) data with measurements $\omega(k)$ for $k = 0,1, \ldots$ as sample times with period T_s



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→ $\theta(2) = \theta(1) + T_s\omega(2) = 0.1° + 0.1s \cdot 2\frac{°}{s} = 0.3°$

- $k = 3$

→ $\theta(3) = \theta(2) + T_s\omega(3) = 0.3° + 0.1s \cdot 3\frac{°}{s} = 0.6°$

3D gyroscope data

- 3D gyroscopes measure the angular rate ω in deg/s (or rad/s) about the *x*, *y*, and *z*-axis
- By integrating the data, the orientation angle of each axis can be determined
- Discrete (sampled) data with measurements $\omega(k)$ for $k = 1,2, \ldots$ as sample times with period T_s

$$\theta(k) = \theta(k-1) + T_s\omega(k)$$

new angle value last angle value angular rate

the inital angle value $\theta(0)$ needs to be known! Easiest to start at $\theta(0) = 0$

Determine a quaternion based on gyroscope data

- The quaternion derivative $\dot{\boldsymbol{q}}$ is given by:

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \circ \boldsymbol{\Omega}$$

Where:

- q represents the 3D-orientation of the body relative to the inertial/world frame
- $\boldsymbol{\Omega} = [0 \quad \boldsymbol{\omega}]$ is the vector quaternion corresponding to the is the angular velocity vector $\boldsymbol{\omega} = [\omega_x \quad \omega_y \quad \omega_z]^T$ expressed in the body frame

- Integrate $\boldsymbol{q}$ over time using the derivative:

$$\begin{aligned}\boldsymbol{q}(k) &= \boldsymbol{q}(k-1) + \dot{\boldsymbol{q}} \cdot T_s \\ &= \boldsymbol{q}(k-1) + \frac{1}{2}\boldsymbol{q}(k-1) \circ \boldsymbol{\Omega}(\mathbf{k}) \cdot T_s\end{aligned}$$

Integration problem

- Real sensor data always contains errors, i.e.:

true angular rate

$$\omega_m(t) = g \cdot \omega(t) + b$$

measured angular rate scale factor offset

- E.g., constant true angular rate given by $\omega(k) = \omega = 4\frac{\text{deg}}{\text{s}}$, $g = 1, b \neq 0, \theta(0) = 0$

$$\boxed{\theta(k) = \theta(k-1) + T_s\omega(k)}$$

$$k = 1: \qquad \theta(1) = 0 + T_s[\omega + b]$$

$$k = 2: \qquad \theta(2) = T_s[\omega + b] + T_s[\omega + b] = 2T_s\omega + 2b$$

$$k = 3: \qquad \theta(3) = T_s[\omega + b] + T_s[\omega + b] + T_s[\omega + b] = 3T_s\omega + 3b$$

$$\vdots \qquad\qquad \vdots$$

$$k = n: \qquad \theta(n) = \underbrace{nT_s\omega}_{\text{true value}} + \underbrace{nb}_{\text{error}}$$

Error increases with time!
So called ***drift*** effect

3D Gyroscope

Integration problem

- Gyroscope offset is influenced by many external effects, including
  - Temperature
  - Humidity
  - Vibrations
  - Aging
- Dependency (e.g. on temperature) is non-linear and cannot be compensated easily
- Therefore, gyroscopes are usually combined with accelerometers for drift compensation of regarding *x*- and *y*-axis orientation, and with magnetometers regarding *z*-axis orientation

Exercise L5.1 – Orientation estimation based on Gyroscope Data

- Implement the numerical integration approach to obtain the orientation estimation for *x*-, *y*- and *z*-axis based on the gyroscope data

- Load the file ‘no_motion.mat’ which contains the measured data of the Mambo Drone when the drone is places statically on the table for about 60s (you will need the loaded variables Ts, time and gyro)

- Use your new function to convert the gyroscope data contained in the mat-file into euler angles and plot the euler angles. What can you observe?

You can use the ex_5_1_template available in ILIAS!

Exercise L5.2 – Simple drift compensation

- Now extend the previous script by a simple drift compensation approach: calculate the mean over the first 100 samples of the gyro data
- Subtract that mean/offset vector from all samples of the gyroscope data before estimating the orientation
- Plot the data. What can you observe?

FYI: no template here, use your file from L5.1

Exercise L5.3 – Quaternions from gyro data

- Implement the quaternion integration approach to obtain quaternions based on the gyroscope data
- load the file ‘no_motion.mat’ which contains the measured data of the Mambo Drone when the drone is places statically on the table for about 60s (you will need the loaded variables Ts, time and gyro)
- Use the euler angles (obtained by conversion of the quaternions into rpy) to plot the euler angles
- Verify that the result is the same as in 5_1

You can use the ex_5_3_template available in ILIAS!